

Multi-Language Audio Retrieval for Code-Switched DRC Swahili Speech

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Abstract

We address the lack of a usable French–Swahili–Lingala code-switched retrieval benchmark for eastern DRC speech technology. In eastern DRC cities such as Goma and Bukavu, many speakers use Swahili as the structural base of everyday speech while fluidly inserting French for official or technical terms and Lingala for emphasis, slang, or social nuance. Our approach is to build a synthetic trilingual corpus, derive a controlled text-to-text retrieval benchmark, and validate the benchmark with a simple lexical baseline before training stronger multilingual or cross-modal models. The resulting corpus contains 9,000 text items and 623 aligned synthetic audio clips. The benchmark contains 164 test queries, each evaluated against 50 candidates with one gold target, nine hard negatives, and forty random negatives. A TF-IDF word-bigram baseline reaches 0.5803 MRR, 0.4085 Recall@1, and 0.6232 nDCG@5. Our main contributions are the benchmark itself, a fully specified evaluation protocol, and an error analysis showing that failures cluster around orthographic near-duplicates, same-template lexical substitutions, and Lingala-bearing switch tokens. The main limitation is that all results are bounded to this synthetic corpus.

1 Introduction

Low-resource speech technology for the Democratic Republic of Congo lacks a usable benchmark for French–Swahili–Lingala code-switched retrieval. This is the specific problem addressed in this paper. We focus on retrieval because it is a practical first task for multilingual speech systems and because a benchmark is necessary before stronger encoders or

audio-text models can be compared fairly.

The problem is important because these languages appear together in real communication contexts, including humanitarian and public-service settings [7]. Prior work on Congolese Swahili speech translation already shows that the region is important and technically challenging [3]. More broadly, recent work on code-switched speech and multilingual retrieval shows that benchmark availability strongly shapes what can be measured and improved [16][6][2].

The problem is hard because naive approaches fail in two different ways. First, there is no public benchmark in the exact French–Swahili–Lingala setting, so model evaluation is not standardized. Second, code-switched retrieval is difficult even when data exists: surface lexical overlap can be misleading, orthographic variation is common, and code-switch points can alter meaning or disrupt matching. A purely lexical model can therefore perform well on some synthetic siblings but still fail on minimally different hard negatives.

The problem has not been solved before because previous resources do not match the target setting. Existing corpora cover Congolese Swahili speech translation [3], multilingual synthetic code-switched evaluation [16], or code-switching in other languages and tasks [15][12]. Supporting resources such as Gamayun, WaxalNLP, Common Voice French, SERENGETI, and AfriBERTa are useful building blocks, but none of them directly provides a French–Swahili–Lingala code-switched retrieval benchmark with frozen evaluation splits [13][14][5][8][11][1]. Our work differs by constructing and documenting that benchmark explicitly.

Our approach is to build a synthetic trilingual corpus, derive a controlled retrieval benchmark from the frozen test split, run a TF-IDF word-bigram baseline,

and analyze the observed failures. The main result is that the benchmark is usable and non-trivial: the baseline reaches 0.5803 MRR, 0.4085 Recall@1, and 0.6232 nDCG@5 on 164 queries, while manual inspection shows systematic failure modes rather than random errors. The main limitations are that the benchmark is synthetic, the switch-band distribution is imbalanced, and source-family overlap across splits must be fixed before training stronger learned models.

1.1 Summary of Contributions

- Section 2 defines the released corpus, benchmark construction procedure, and evaluation protocol.
- Section 3 establishes a reproducible TF-IDF retrieval baseline for the benchmark.
- Section 4 reports benchmark results by overall score, switch band, and Lingala presence.
- Section 5 provides a manual error analysis with representative examples and recurring failure categories.
- Section 6 states the conclusions and limits the claims to the observed synthetic dataset.

2 Data and Benchmark

2.1 Corpus Construction

The corpus was created from parallel French–Swahili and French–Lingala sentence sources, then expanded with rule-based code-switching and copy-switch variation. The generator follows the Matrix Language Frame intuition: one language provides the grammatical frame, while embedded spans come from the other languages [9]. In our corpus, Swahili is usually the matrix language.

An example from the released text set is: *Leo tuko na probleme ya maji, mairie ilisema travaux zinaanza kesho matin*. The sentence keeps a Swahili frame while inserting French content words. This is the type of mixed-language input that the benchmark is meant to test.

Table 1 summarizes the released resources. The text set contains 9,000 items. The aligned audio subset contains 623 synthetic clips. The audio is useful

Table 1: Released corpus and benchmark summary.

Item	Value
Text records	9,000
Text split	7,200 / 900 / 900
Audio clips	623
Audio split	498 / 62 / 63
Benchmark queries	164
Candidates per query	50

for future cross-modal experiments, but this report evaluates only text-to-text retrieval because the text benchmark is the most stable part of the pipeline at this stage.

2.2 Benchmark Construction

The retrieval benchmark is derived from the text test split. We keep only test items that have a valid paired target from the same synthesis lineage, which yields 164 queries rather than all 900 test examples. Each query is evaluated against a fixed candidate pool of size 50: one gold target, nine hard negatives, and forty random negatives.

This controlled setup follows the logic of benchmark-driven evaluation used in recent low-resource retrieval work. It lets us test whether a model can recover the intended paired item under hard distractors instead of rewarding trivial exact-match retrieval.

The benchmark is imbalanced by switch band. Out of 164 queries, 154 are low-switch, 7 are medium-switch, and 3 are high-switch. This matters for interpretation: any high-switch conclusion is based on only three queries. The benchmark also contains 34 Lingala-bearing queries and 130 queries without Lingala.

One important limitation emerged during audit. The exported `text_leakage_report.csv` shows that 1,472 source families appear in more than one split, including 138 families that appear in train, dev, and test. The TF-IDF baseline reported here does not train on the train split, so this overlap does not inflate the current scores. However, future learned models should use a family-disjoint split.

2.3 Metrics

We report Mean Reciprocal Rank (MRR), Recall@1, and nDCG@5. MRR is the primary metric because it measures how early the correct item appears. Recall@1 captures whether the first result is correct. nDCG@5 summarizes the quality of the top of the ranked list. We also compute a simple code-switch robustness score:

$$\text{CSRS} = \frac{\text{MRR}_{\text{high}}}{\text{MRR}_{\text{low}}}. \quad (1)$$

The evaluation code also includes paired bootstrap resampling for ranking metrics and McNemar’s test for Recall@1. In this report, however, only one actual model was run. The only statistical test we can report honestly is the self-comparison sanity check, which gives zero deltas and $p = 1.0$ for every metric. We therefore present descriptive results only, not a meaningful model-versus-model significance claim.

3 Baseline Model

We use a TF-IDF word-bigram retriever. This is a weak but useful baseline. It checks whether the benchmark construction, candidate pools, ranking code, and exported metrics behave correctly before we invest in multilingual encoders or audio-text models.

This choice is also intentionally conservative. The synthetic corpus contains many near-paraphrases from the same generation lineage, so a lexical baseline should perform above random if the benchmark is well formed. At the same time, strong errors on hard negatives are still informative because they reveal where lexical overlap is insufficient.

4 Results

Table 2 gives the overall result. The baseline retrieves the gold item at rank 1 for 67 of the 164 queries. The overall MRR is 0.5803, which shows that the correct item is often near the top but far from guaranteed to be first.

Figure 1 breaks performance down by switch band. The raw MRR values are 0.5837 for low-switch, 0.3976 for medium-switch, and 0.8333 for high-switch. The high-switch result looks best, but it is not the most reliable result because it comes from

Table 2: TF-IDF baseline results on the 164-query benchmark.

Metric	Score
MRR	0.5803
Recall@1	0.4085
nDCG@5	0.6232
CSRS	1.4277

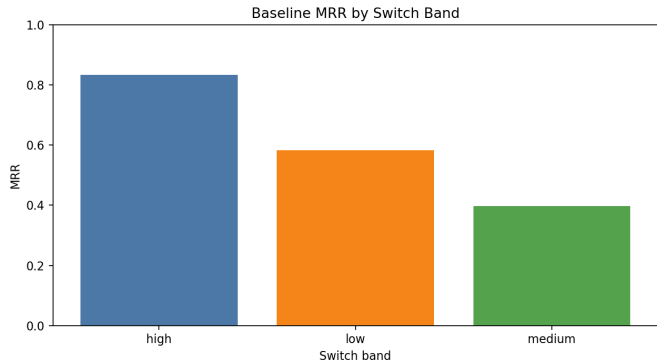


Figure 1: MRR by switch band. The high-switch score should not be overinterpreted because that subset contains only 3 queries.

only three queries. The more stable pattern is the drop on the medium-switch subset, where Recall@1 falls to 0.1429. On this benchmark, medium-switch items are the hardest condition.

Figure 2 shows a smaller but consistent gap by Lingala presence. Queries without Lingala reach 0.5912 MRR and 0.4154 Recall@1. Queries with Lingala drop to 0.5385 MRR and 0.3824 Recall@1. This suggests that the baseline is less stable when Lingala tokens appear, which is plausible given the smaller amount of Lingala-bearing material and the lexical nature of the model.

5 Error Analysis

We manually inspected the 20 worst failed queries, using the exported per-query ranks and the paired gold texts. Because the current CSV export stores rank metrics but not the top-ranked distractor text, this analysis focuses on what makes the gold pair hard rather than on the full ranked list. Even with that limitation, the failure patterns are clear.

We observed three recurring categories. First, 7 of the 20 failures are orthographic near-duplicates,

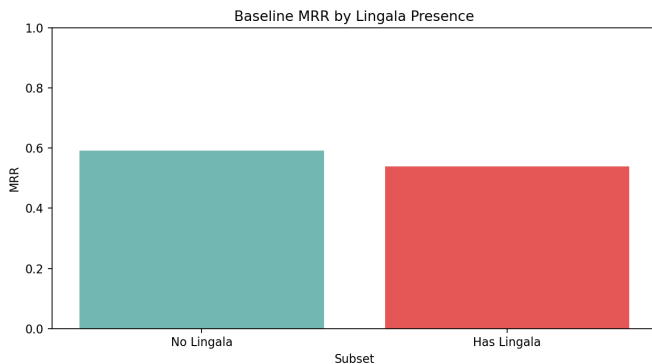


Figure 2: Lingala-bearing queries are slightly harder for the lexical baseline.

Table 3: Representative failure cases from manual inspection.

Query		Gold	
<i>Wananchi wanataka usalama na barabara routes.</i>		<i>Wananchi wanatakae usalama na barabara routese.</i>	
Category: orthographic near-duplicate.			
<i>Serikali bazali mpango programme wa ajira kwa vijana.</i>		<i>Serikali ilitangaza mpango programmee wa ajira ndenge vijana.</i>	
Category: same-template lexical substitution.			
<i>budget ya afya imeongezeka mwaka huu.</i>		<i>budget ya afya bazali soki huu.</i>	
Category: Lingala-bearing switch-token mismatch.			

where the query and gold differ by light suffix noise such as *wanataka* versus *wanatakae* or *routes* versus *routese*. Second, 5 failures are same-template lexical substitutions, where the overall topic is preserved but one or two key words are replaced. Third, 8 failures involve Lingala-bearing or switch-token substitutions such as *soki*, *batu*, or *bazali*. These tokens often preserve surface overlap while changing which paired item is correct.

These errors show that the benchmark is not trivial. The model is not simply failing on unrelated negatives. It often faces highly similar candidates from the same synthetic template family and cannot reliably separate the intended pair from minimal orthographic or lexical changes. This is exactly the type of behavior we wanted the first benchmark to expose.

6 Discussion

Two conclusions are supported by the current evidence. First, the project now has a usable benchmark: the dataset is frozen, the candidate pools are fixed, the evaluation metrics are exported, and the baseline produces interpretable successes and failures. Second, the reported scores are properties of this synthetic benchmark, not evidence about spontaneous DRC speech.

The benchmark is still early-stage and its limitations matter. The switch-band imbalance is severe, the audio subset is small, and the source-family overlap across splits must be fixed before training stronger learned models. The current results are therefore best understood as benchmark-validation results, not as evidence that the retrieval problem is solved for eastern DRC speech.

7 Future Work

The most important next step is to obtain a proper local voice dataset from eastern DRC. We already prepared the community-collection path because a realistic benchmark will eventually require real speakers from places such as Goma and Bukavu, not only synthetic renderings. That future dataset should better capture the fluid way local speakers move across Swahili, French, and Lingala within a single utterance.

The next experimental phase will follow the broader plan already defined for the project. On the text side, we plan to evaluate SERENGETI as the main multilingual African-language encoder and AfriBERTa as a low-resource African-language text baseline [11][1]. On the cross-modal side, we plan to fine-tune a CLAP-style architecture to align audio and text representations and to compare that setup with a CLASP-style alternative for multilingual audio-text retrieval [4][10]. These experiments will cover text-to-text, text-to-audio, and audio-to-text retrieval.

We will keep MRR as the primary ranking metric and continue to report Recall@1, nDCG@5, and the code-switch robustness score. Once multiple real models are run, we will use the existing paired bootstrap and McNemar testing code to support actual between-model significance claims rather than de-

scriptive single-model reporting.

8 Conclusion

This report turns the project from an idea into an experimental asset. We produced a documented synthetic French–Swahili–Lingala corpus, derived a controlled retrieval benchmark from it, and showed that even a simple lexical baseline yields informative successes and failures. The benchmark now gives us a concrete starting point for stronger multilingual and cross-modal retrieval experiments.

The main value is not the TF-IDF score by itself. The main value is that we now have a reproducible benchmark, explicit limitations, and a clear next path toward real eastern DRC voice data and stronger retrieval models.

References

- [1] Castorini. Afriberta large. Hugging Face model, 2026. Available at: https://huggingface.co/castorini/afriberta_large. Accessed: 2026-04-14.
- [2] Jian Cui, Brian Kingsbury, Bhuvana Ramabhadran, George Saon, Tom Sercu, Kartik Audhkhasi, Abhinav Sethy, Khe Chai Sim, and Melissa Nussbaum-Thom. Multilingual representations for low resource speech recognition and keyword search. In *2015 IEEE Workshop on Automatic Speech Recognition and Understanding (ASRU)*, pages 259–266, Scottsdale, AZ, USA, 2015.
- [3] Pavel Denisov, Manuel Mager, and Ngoc Thang Vu. IMS’ systems for the IWSLT 2021 low-resource speech translation task. *arXiv preprint arXiv:2106.16055*, 2021.
- [4] Benjamin Elizalde, Shah Zarar, and Bhiksha Raj. Cross modal audio search and retrieval with joint embeddings based on text and audio. In *ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 4095–4099, Brighton, UK, 2019.
- [5] Google. Waxalnlp. Hugging Face dataset, 2026. Available at: <https://huggingface.co/datasets/google/Waxalnlp>. Accessed: 2026-04-14.
- [6] Léopold Hillah, Mateusz Dubiel, and Luis A. Leiva. “Te vienes? Sure!” Joint Fine-tuning of Language Detection and Transcription Improves Automatic Recognition of Code-Switching Speech. In *ACM Conversational User Interfaces 2024 (CUI ’24)*, pages 1–11, Luxembourg, Luxembourg, 2024. ACM.
- [7] Ussen Kimanuka, Ciira wa Maina, Osman Buyuk, and Masika Kassay Godelive. A congolese swahili task-oriented dialogue system for addressing humanitarian crises. *Engineering, Technology & Applied Science Research*, 15(5):27387–27399, 2025.
- [8] Mozilla Data Collective. Common voice scripted speech 25.0 – french. Dataset page, 2026. Available at: <https://mozilladatacollective.com/datasets/cmn5zugst00w3nv07upovf2bg>. Accessed: 2026-04-14.
- [9] Carol Myers-Scotton. *Duelling Languages: Grammatical Structure in Codeswitching*. Oxford University Press, 1993.
- [10] O. O. Nuga et al. Multi-lingual contextual audio transcription and translation to selected native languages. In *2025 2nd Asia Pacific Conference on Innovation in Technology (APCIT)*, pages 1–6, Mysore, India, 2025.
- [11] Salomey Osei, David Ifeoluwa Adelani, Jesujoba Alabi, et al. Serengeti: Massively multilingual language models for africa. *Findings of the Association for Computational Linguistics: ACL 2023*, 2023. Available at: <https://arxiv.org/abs/2212.10785>.
- [12] K. S. Sailaja. Switchlang ai: Advanced nlp for seamless code-switching & multilingual text processing. *International Journal of Scientific Research in Engineering and Management*, 2025.
- [13] Translators without Borders. Gamayun congolese swahili–french kit. Download page, 2026. Available at: <https://gamayun.translatorswb.org/download/gamayun-medium-kit-15k-chunk-1-2-swc-fra/>. Accessed: 2026-04-14.

- [14] Translators without Borders. Gamayun language resources portal. Project website, 2026. Available at: <https://gamayun.translatorswb.org/data/>. Accessed: 2026-04-14.
- [15] Peng Xie, Xingyuan Liu, Tsz Wai Chan, Yequan Bie, Yangqiu Song, Yang Wang, Hao Chen, and Kani Chen. Switchlingua: The first large-scale multilingual and multi-ethnic code-switching dataset. *arXiv preprint arXiv:2506.00087*, 2025.
- [16] Brian Yan, Injy Hamed, Shuichiro Shimizu, Vassista Lodagala, William Chen, Olga Iakovenko, Bashar Talafha, Amir Hussein, Alexander Polok, Calvin Chang, Dominik Klement, Sara Althubaiti, Puyuan Peng, Matthew Wiesner, Tamar Solorio, Ahmed Ali, Sanjeev Khudanpur, Shinji Watanabe, Chih-Chen Chen, Zhen Wu, Karim Benharrak, Anuj Diwan, Samuele Cornell, Eunjung Yeo, Kwanghee Choi, Carlos Carvalho, and Karen Rosero. CS-FLEURS: A massively multilingual and code-switched speech dataset. *arXiv preprint arXiv:2509.14161*, 2025.

A Contribution Statement

For this report, many tasks were done collaboratively. We jointly designed the project scope, defined the corpus construction strategy, decided the benchmark protocol, interpreted the results, and wrote the main narrative of the paper together.

Athanase led the report revision, benchmark framing, and the final error-analysis write-up. Stephen led the corpus generation pipeline, data organization, and the supporting engineering around the collection workflow and released artifacts. The results section reflects shared analysis, but the final wording and paper editing for this version were completed jointly and then consolidated by Athanase.